ZIDIO INTERNSHIP PROJECT | PROJECT REPORT

RetailPulse

AI-Powered Customer Analytics & Demand Forecasting

Transforming Retail Intelligence with AI-Driven Insights.

DEVELOPED BY

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PROJECT DURATION

21/09/2026 – 20/10/2026

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1. Project Overview

RetailPulse is an AI-powered customer analytics and demand forecasting project developed as part of the Zidio Internship. It turns sales, customer, product and inventory data into business insights using machine learning and interactive visualization, and it is built on synthetic retail datasets.

The project combines customer segmentation, demand forecasting and inventory analytics in an interactive dashboard built with Python and Streamlit.

1.1 Project Details

Item	Details
Project title	RetailPulse: AI-Powered Customer Analytics & Demand Forecasting
Tagline	Transforming Retail Intelligence with AI-Driven Insights.
Programme	Zidio Internship Project
Team members	Gautam Rishi, Prerna, Riddhima Singh, Asiya Sadhika.M
Project duration	21/09/2026 – 20/10/2026
Data	Synthetic retail datasets (sales, customer, product and inventory data)

1.2 Vision and Objectives

The project is guided by the following objectives:

- Analyze sales performance and revenue trends.
- Segment customers with RFM analysis.
- Visualize demand trends and sales forecasts.
- Monitor inventory levels and stock issues.

1.3 Target Users and Use Cases

RetailPulse is intended for the following users:

- Retail store managers
- E-commerce businesses
- Inventory and supply chain teams
- Business analysts

1.4 Business Value

The project delivers the following business value:

- A clearer view of sales and revenue performance.
- Better insight into customer purchasing behavior.
- Stronger demand planning and forecasting.
- Less manual reporting, with centralized analytics.

2. System Architecture

RetailPulse follows an end-to-end AI and MLOps workflow in which raw sales data becomes forecasts, inventory decisions, and continuously monitored models. Figure 1 presents the workflow, which consists of eight stages.

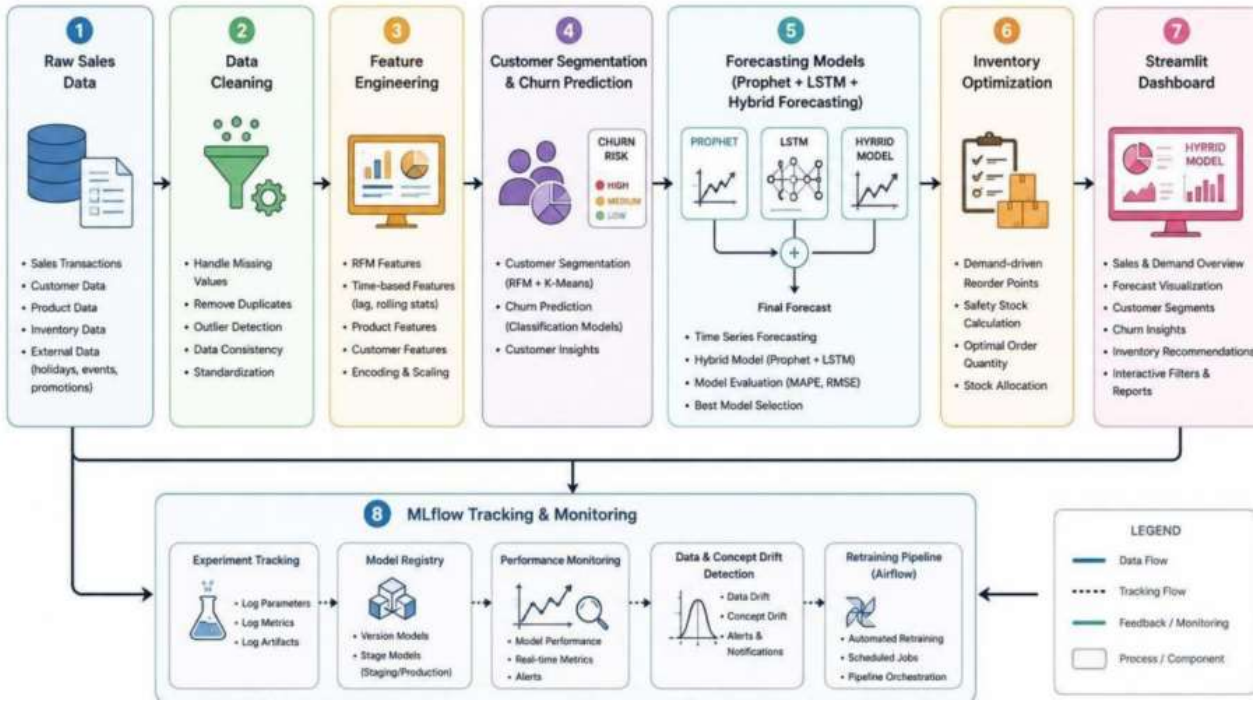


Figure 1: End-to-end AI and MLOps workflow of RetailPulse

2.1 Workflow Stages

#	Stage	Components
1	Raw Sales Data	Sales transactions, customer data, product data, inventory data, external data (holidays, events, promotions)
2	Data Cleaning	Handle missing values, remove duplicates, outlier detection, data consistency, standardization
3	Feature Engineering	RFM features, time-based features (lag, rolling stats), product features, customer features, encoding and scaling
4	Customer Segmentation & Churn Prediction	Customer segmentation (RFM + K-Means), churn prediction (classification models), customer insights; churn risk levels: high, medium, low
5	Forecasting Models (Prophet + LSTM + Hybrid Forecasting)	Time series forecasting, hybrid model (Prophet + LSTM), model evaluation (MAPE, RMSE), best model selection, producing the final forecast
6	Inventory Optimization	Demand-driven reorder points, safety stock calculation, optimal order quantity, stock allocation
7	Streamlit Dashboard	Sales and demand overview, forecast visualization, customer segments, churn insights, inventory recommendations, interactive filters and reports

#	Stage	Components
8	MLflow Tracking & Monitoring	Experiment tracking (log parameters, metrics, artifacts); model registry (version models, stage models: staging/production); performance monitoring (model performance, real-time metrics, alerts); data and concept drift detection (data drift, concept drift, alerts and notifications); retraining pipeline with Airflow (automated retraining, scheduled jobs, pipeline orchestration)

3. Project Timeline

The project was delivered over four weeks, from requirements to deployment.

Week	Period	Activities
Week 1	20 Sep – 27 2026	Planning, dataset collection and preprocessing, EDA
Week 2	28 Sep – 4 Oct 2026	Feature engineering, RFM segmentation, churn model
Week 3	5 Oct – 11 Oct 2026	Prophet and LSTM forecasting, hybrid model, inventory optimization
Week 4	12 Oct – 20 Oct 2026	Streamlit dashboard, Docker, deployment and testing, final report

4. Key Features and Acceptance Criteria

RetailPulse comprises eight features, each with a clear acceptance test.

ID	Feature	Description	Acceptance Criteria
F1	Customer Segmentation	Groups customers based on behavior	Customers categorized successfully
F2	Churn Prediction	Predicts at-risk customers	Accurate churn insights generated
F3	Demand Forecasting	Predicts future demand	Reliable forecast trends shown
F4	Hybrid Forecasting	Combines multiple models	Improved prediction quality
F5	Inventory Optimization	Optimizes stock levels	Reduced overstocking risk
F6	Dashboard Analytics	Interactive visualization	Real-time insights accessible
F7	Drift Detection	Detects data changes	Alerts for model degradation
F8	MLflow Tracking	Tracks experiments	Performance history maintained

5. Technology Stack, Monitoring and Deployment

Category	Technologies / Approach
Stack	Python, Streamlit, Scikit-learn, TensorFlow, Prophet, LSTM, MLflow, Optuna, Airflow, Docker, Git & GitHub
Monitoring	Drift detection, model performance tracking, MLflow experiments
Deployment	Streamlit Cloud hosting, Docker containers, GitHub version control and CI/CD

The conclusion of the project additionally records that the application is deployed using Docker and Render, making it accessible online. The dashboard was developed using Python, Streamlit, and Plotly.

6. Modeling, MLOps, Challenges and Solutions

Hybrid modeling and MLOps practices addressed the hard parts of the project.

Area	Focus	Details
Modeling	Multiple views of the customer and the future	RFM and clustering segmentation, ML churn prediction, Prophet + LSTM hybrid forecasting.
MLOps	Tracking, tuning, detection, orchestration	MLflow tracking, Optuna tuning, drift detection, Airflow orchestration.
Challenges	The hard parts were operational	Inconsistent sales data, integrating multiple forecasting approaches, deployment configuration, Docker optimization.
Solutions	A modular path from data to decisions	Preprocessing pipeline, hybrid approach, modular architecture, automated workflows.

7. Dashboard and Results

The dashboard forecasts demand and recommends stock. The screens include the home screen, the forecast trend with weekly rhythm (Saturday peak), forecast versus adjusted demand (average 28,461) and inventory recommendations (average stock 389,789, minimum 319,939).

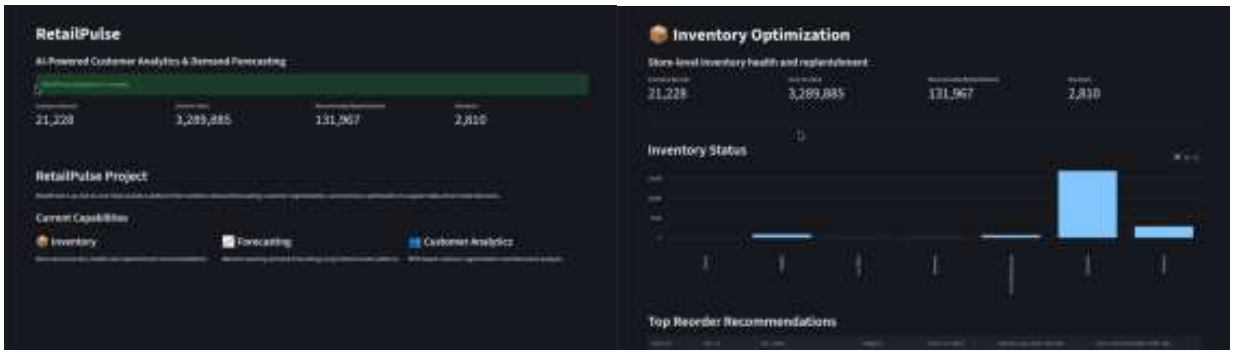


Figure 2: Dashboard home screen (left) and inventory dashboard (right)



Figure 3: Forecast dashboard (left) and customer segmentation (right)

7.1 Model Insights

Recency drives churn, and drift alerts keep the models honest. The following charts summarize the demand analysis.

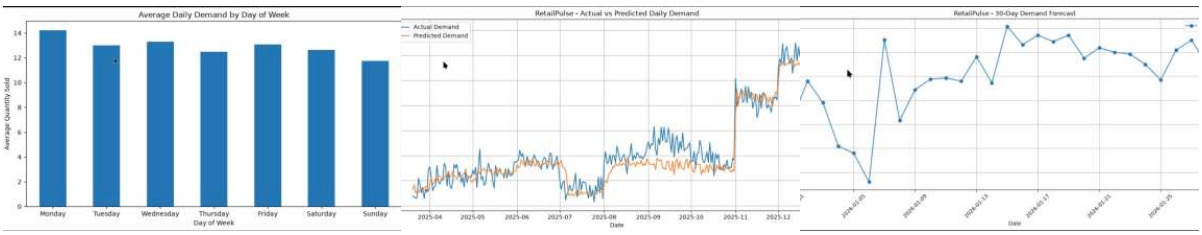


Figure 4: Average daily demand by day of week; actual vs predicted daily demand; 30-day demand forecast

7.2 Interpretation of Results

Average Daily Demand by Day of Week. Average daily demand is highest on Monday at around 14.2 units, while Sunday records the lowest demand at approximately 11.7 units. Demand remains relatively consistent throughout the week, with a slight decline toward the weekend, which can help retailers plan inventory and stock replenishment.

Actual vs Predicted Daily Demand. The graph shows that predicted demand generally follows actual demand trends, capturing major fluctuations and seasonal changes throughout the year. However, the model underestimates some demand peaks, especially in November and December, indicating opportunities to improve forecasting accuracy.

30 Day Demand Forecast. The 30-day demand forecast predicts daily retail demand of approximately 11,080 to 11,400 units, showing moderate fluctuations throughout January 2026. These forecasts help retailers anticipate future demand, plan inventory levels, and make better stock replenishment decisions.

8. Learnings, Conclusion and Roadmap

8.1 Learnings

The project provided hands-on experience with ML workflows, end-to-end deployment, Streamlit dashboards, forecasting and churn analysis. Best practices applied include modular code, GitHub, MLflow tracking and Dockerized deployment.

8.2 Conclusion

RetailPulse demonstrates how data analytics can help retail businesses make better decisions by analyzing sales performance, customer behavior, demand patterns, and inventory levels. The project combines data analysis, RFM-based customer segmentation, demand forecasting visualizations, and inventory analytics in an interactive dashboard developed using Python, Streamlit, and Plotly.

The application is deployed using Docker and Render, making it accessible online. Overall, this project provided practical experience in data analysis, dashboard development, and deployment while demonstrating how data-driven insights can support better retail planning, customer understanding, and inventory management.

8.3 Roadmap

- Real-time analytics
- Cloud-native deployment
- Recommendation systems
- Enhanced automation and monitoring

Project Team: Gautam Rishi, Prerna, Riddhima Singh, Asiya Sadhika.M